

Navigating Truth in Multimodal Fact-checking via Retrieval- and Reasoning-Enhanced Large Language Models

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Abstract

Recent studies show that claims incorporating both text and images spread more effectively than those with text alone, presenting significant challenges for multimodal fact-checking. The rapid development of Multi-modal Large Language Models (MLLMs) has greatly advanced research in this field, enabling stronger performance. However, existing MLLM-based fact-checking methods fail to fully exploit visual evidence, and their reliance on rigid fine-tuning templates limits context-aware explanations and leads to weak deep reasoning. To address these limitations, we propose FACTCOMPASS, a novel framework that combines reasoning-aware fine-tuning with large-scale rule-based reinforcement learning and incorporates a semantic- and knowledge-enhanced retrieval module to strengthen deep reasoning and improve evidence utilization. This framework enhances evidence retrieval by obtaining semantically relevant evidence images, enriching the contextual understanding of claim-related images, and refining textual evidence at the knowledge level. To further enhance reasoning, we introduce a self-refining reinforcement fine-tuning strategy: (1) distilling GPT-4o's reasoning from partially fact-checking data for cold-start Chain-of-Thought learning; (2) activating reasoning across broader datasets using prior

knowledge and rejection sampling; (3) applying Group Relative Policy Optimization to explore diverse reasoning paths and optimize factual consistency. Extensive experiments have demonstrated the effectiveness of the proposed framework.

CCS Concepts

• Information systems → Social networks; Multimedia information systems.

Keywords

Multimodal fact-checking, Deep Reasoning, Retrieval-augmented generation

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1 Introduction

The unregulated spread of false information can distort public opinion, manipulate markets, and even endanger public health. Fact-checking, which assesses the veracity of written and spoken claims by referencing supporting evidence [14, 38, 47, 50], is an effective means of identifying falsehoods and has become a key tool for ensuring information credibility. As social media and generative AI accelerate the dissemination of information at an unprecedented pace, manual fact-checking has become insufficient to meet the

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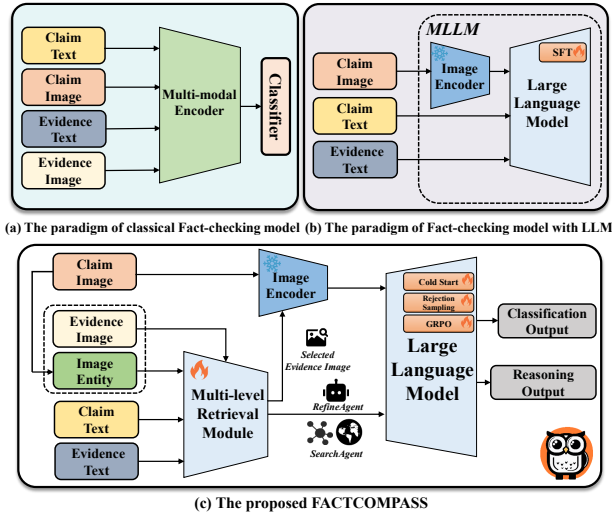


Figure 1: FACTCOMPASS enhances both textual and visual evidence retrieval and adopts a more sophisticated training strategy to stimulate reasoning capabilities.

challenge. Therefore, to address the demands of scale, automated fact-checking is becoming indispensable.

Information is conveyed through multimodal media, primarily text and images. Existing text-based fact-checking methods have achieved significant progress. However, with the rise of multimedia technologies, an increasing number of rumor spreaders are leveraging combinations of text and images to attract attention and spread multimodal false information [35, 44]. Common types of multimodal misinformation can be categorized into two groups: one involves fabricated text or images [23, 34], and the other [1, 39] involves authentic images presented in new but misleading or incorrect contexts—a phenomenon often referred to as “out-of-context”. Unlike unimodal text-based fact-checking, text and images are inherently heterogeneous, making it challenging to detect false information from such entangled multimodal content.

Early methods [6, 12, 49] fuse multimodal features from claims and evidence using multimodal encoders, followed by a fact-checking classifier to make predictions (as illustrated in Figure 1(a)). These approaches typically adopt concatenation operations or attention mechanisms to capture the underlying semantic relationships between the claims. However, they often struggle to fully comprehend the semantics of news content. In addition, due to inherent model limitations, such methods generally fail to generate convincing and interpretable fact-checking outcomes. In recent years, multimodal large language models (MLLMs) have made significant strides in integrating visual and textual information. Building upon this, several studies [29] have fine-tuned large language models specifically for fact-checking tasks (as shown in Figure 1(b)), where multimodal claims and evidence are organized into supervised fine-tuning (SFT) datasets to train open-source MLLMs. Nevertheless, these methods still face several key challenges: (1) they fail to fully utilize image evidence, overlooking the learning and fine-grained alignment of visual cues; (2) they rely on rigid processing pipelines that are

error-prone and lack robust deep reasoning capabilities, ultimately constraining the model’s ability to understand complex and realistic news contexts.

To address the limitations of existing methods in image utilization and reasoning capabilities, we propose FACTCOMPASS, a novel framework that combines multi-level retrieval modules with a self-optimizing reinforcement fine-tuning strategy to advance the development of multimodal fact-checking. The framework is specifically designed to overcome two core challenges in current approaches: (1) To address the insufficient use of image evidence and the lack of fine-grained visual alignment, FACTCOMPASS introduces a Semantic-enhanced Retrieval Module, which constructs a dedicated claim-evidence image pair dataset and adopts a Fact NCE loss function to learn a unified feature space between claims and evidence images. This enables accurate matching of semantically relevant image evidence, thereby enhancing the model’s ability to understand and reason over visual information. Additionally, a Knowledge-enhanced Retrieval Module is well-designed to incorporate knowledge graphs and structured knowledge bases (e.g., Wikidata) via SearchAgent and RefineAgent, supplementing and optimizing textual evidence to enhance contextual understanding and retrieval efficiency. (2) To tackle rigid processing pipelines and weak reasoning capabilities, FACTCOMPASS proposes a reinforcement learning method named Group Relative Policy Optimization (GRPO) to replace traditional template-based reasoning. This approach encourages the model to explore diverse and deeper reasoning paths through trial-and-error.

The overall multimodal fact-checking process of FACTCOMPASS is illustrated in Figure 1(c), consisting of three major components: (1) Semantic-enhanced Retrieval Module: Enhances semantic matching of image evidence through multi-source image retrieval and alignment in a unified feature space. (2) Knowledge-enhanced Retrieval Module: Expands the context of textual evidence using named entity recognition and knowledge graph linking. The SearchAgent retrieves relevant knowledge, while the RefineAgent refines the results to improve information quality. (3) Large Language Model Training Pipeline: Composed of three stages: 1) Cold-Start: Establishes a reasoning foundation by leveraging GPT-4o to generate initial reasoning data. 2) Rejection Sampling: Generates and filters long-chain reasoning trajectories to reinforce accurate patterns. 3) GRPO: Uses reinforcement learning to optimize reasoning strategies, promoting active reasoning evolution beyond traditional supervised fine-tuning. Through the synergy of these modules, FACTCOMPASS forms a closed-loop system in image understanding, textual refinement, and reasoning evolution, effectively improving the accuracy, flexibility, and interpretability of multimodal fact-checking.

Our contributions are summarized as follows. (1) We propose FACTCOMPASS, a novel multimodal fact-checking framework that integrates multi-level retrieval modules and a self-refining reinforcement fine-tuning strategy to jointly improve image evidence utilization and reasoning capabilities. (2) We design a semantic-enhanced retrieval module with a dedicated claim-image dataset and a Fact NCE loss to learn a unified feature space, enabling fine-grained alignment between claims and image evidence. (3) We introduce group relative policy optimization, a reinforcement learning-based

reasoning method that replaces rigid template-based pipelines by encouraging diverse and deeper reasoning through trial-and-error.

2 Related Work

2.1 Multi-modal Fact-checking.

Prior research on multimodal fact-checking has primarily focused on textual analysis, with limited utilization of visual information. Early approaches, such as Jaiswal et al. [17], matched random image-title pairs, while Sabir et al. [31] advanced this by swapping named entities to generate more challenging mismatches. Khattar et al. [18] introduced the Multimodal Variational Autoencoder (MVAE), which combines a bimodal autoencoder with a classifier to learn shared text-visual representations. However, many fabricated examples in early datasets were overly simplistic or linguistically biased, making them easily detectable by basic models [22]. To address these limitations, Luo et al. [22] developed the NewsCLIPpings dataset, while datasets such as Factify [28] further enriched multimodal misinformation detection with more diverse and realistic samples. Recent advancements have emphasized cross-modal consistency and reasoning. Sahar et al. [1] introduced a consistency-checking network (CCN) to align visual and textual modalities. Dhankar et al. [10] applied cosine similarity between text and visual embeddings, while Du et al. [11] fused DeBERTa and SwinV2 embeddings using co-attention. More recent contributions include Hu et al. [15], who constructed the MR^2 dataset for multimodal reasoning, and the creation of MOCHEG [45], a large-scale fact-checking dataset containing 21,184 annotated claims for training robust multimodal models. AVeriTeC [32] further advances this direction by providing real-world multimodal evidence retrieved from the web for claim verification. DEFAME [5] adopts a six-stage inference-time RAG pipeline with a frozen MLLM, demonstrating strong generalization and factual reasoning capabilities.

2.2 Multi-modal Large language Models.

Recent Multi-modal Large language Models have made remarkable progress in visual perception and cross-modal understanding. BLIP-2 [19] pioneered image-text feature fusion via LLM integration, while LLaVA [21] and Mini-GPT [51] further advanced contextual alignment through instruction tuning. Models such as QwenVL [3], InternVL [7], and YiVL [46] enhanced multimodal reasoning through large-scale pre-training and adaptive fusion mechanisms. Meanwhile, Gemini [37] and SNIFFER [29] demonstrated strong capabilities in misinformation detection and fact verification tasks. With the emergence of advanced LLMs like OpenAI o1 [16] and DeepSeek R1 [13], MLLMs are increasingly capable of handling complex visual reasoning, leveraging methods such as visual chain-of-thought reasoning [33, 40] and retrieval-augmented reasoning [42] to improve interpretability and factual consistency.

3 Methodology

3.1 Architecture Overview

The architecture of FACTCOMPASS, shown in Figure 2 and Figure 3, comprises three core and interrelated components: the Semantic-enhanced Retrieval Module, the Knowledge-enhanced Retrieval

Module, and the LLM module. To further enhance reasoning capability and overall robustness, we introduce a three-stage self-refining reinforcement fine-tuning approach to effectively train the LLM module.

3.2 Semantic-enhanced Retrieval Module

Since the original CLIP architecture was pre-trained on natural image datasets, it lacks support for semantic alignment within the news domain and for dual image-to-image and image-to-text matching. Our goal is to develop a specialized claim-evidence image retrieval module that selects evidence images (along with their corresponding captions) most semantically aligned with the claim image. This approach enables precise matching and improves the model’s ability to understand news facts for fact-checking. During training, we retain the original news captions for each evidence image in the retrieval set, sourced from online documents, to ensure semantic and contextual consistency.

Module Detail. Given CLIP’s outstanding performance in zero-shot perception, which enables effective matching and distinction of general-domain natural images, we built our retrieval module based on the CLIP model, as illustrated in Figure 2. In this setup, CLIP’s image and text encoders are frozen to preserve their pre-trained knowledge. Building on this foundation, we introduce two MLP layers adapter structures after the encoders to fully leverage CLIP’s potential in the domain of image matching for news content. For multimodal claims, we obtain the encoded claim text features t_i^c and image features v_i^c . For evidence image and the corresponding captions, we also obtain the encoded textual features t_i^e and image features v_i^e .

Claim-Evidence Image Pair Generation. To train the retrieval module for fact-checking, we extract a large set of images and their corresponding captions from unmodified visual news documents. First, we classify the images based on their source domain, grouping claim-evidence image-caption pairs from the same document. Next, we compute claim-evidence text similarity scores and select the image most similar to the claim text as the matching evidence image. After manual validation and filtering, we curate a final set of 20,000 claim-evidence image pairs.

Training and Inference. To train our retrieval module, similar to previous works [30, 43], we introduce a dual Fact NCE contrastive loss. We use two types of positive sample pairs during training: 1) Claim image-text pairs, denoted as (v_i^c, t_i^c) , and Evidence image-text pairs, (v_i^e, t_i^e) . 2) Cross-image-text pairs for aligning news semantics, namely, (v_i^c, t_i^e) and (v_i^e, t_i^c) , since t_i^c and t_i^e represent similar events in the news domain. Assuming each mini-batch contains N samples, the image-to-text NCE loss is computed as:

$$\mathcal{L}_{\text{t2t}} = - \sum_{i=1}^N \log \frac{\sum_{(v_i, u) \in \mathcal{M}_i} \exp(\langle v_i, u \rangle / \tau)}{\sum_{j=1}^N \exp(\langle v_i, t_j \rangle / \tau)} \quad (1)$$

where the set of positive pairs $\mathcal{M}_i$ is defined as

$$\mathcal{M}_i = \{(v_i^c, t_i^c), (v_i^e, t_i^e)\} \cup \{(v_i^c, t_i^e), (v_i^e, t_i^c)\}$$

τ represents the temperature coefficient. Similarly, the text-to-image NCE loss is formulated as:

$$\mathcal{L}_{\text{t2i}} = - \sum_{i=1}^N \log \frac{\sum_{(t_i, z) \in \mathcal{Q}_i} \exp(\langle t_i, z \rangle / \tau)}{\sum_{j=1}^N \exp(\langle t_i, v_j \rangle / \tau)} \quad (2)$$

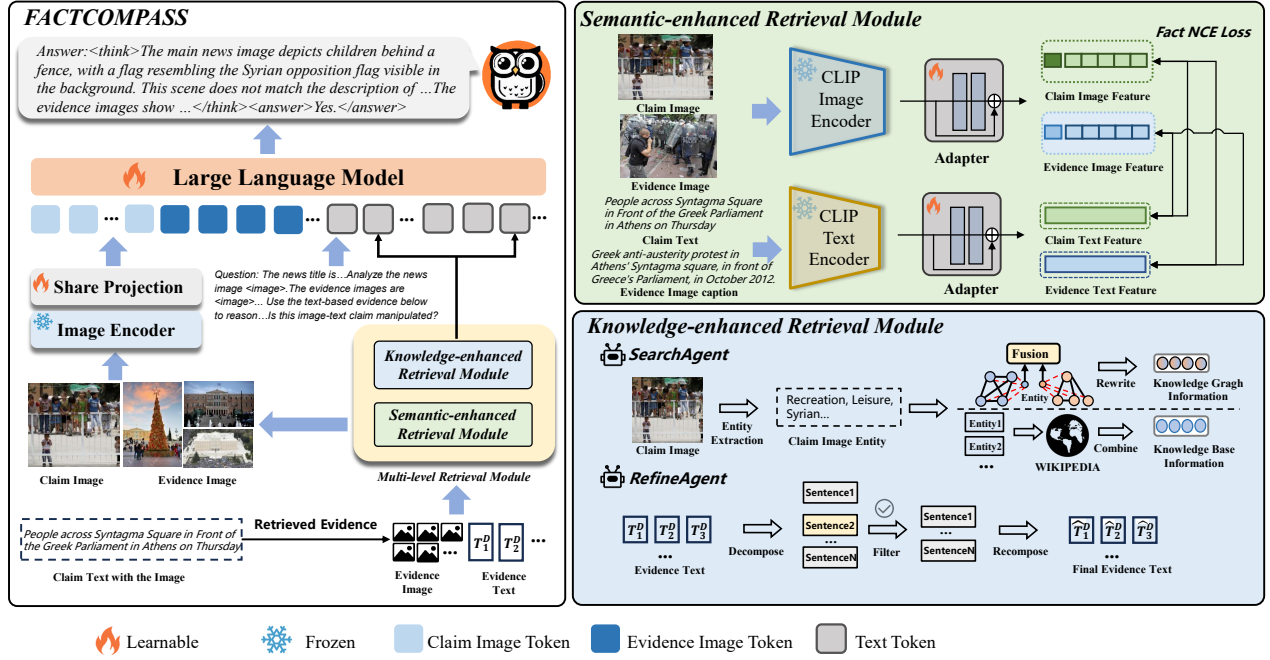


Figure 2: The overall architecture of the proposed FACTCOMPASS.

The total Fact NCE loss is the sum of the image-to-text and text-to-image losses. By modifying the CLIP architecture and training with dual Fact NCE losses, we enhance CLIP’s ability to match images and text in the news domain, while also aligning the feature space of image pairs that describe the same news event. During inference, for a given claim image and a set of candidate evidence images with corresponding captions, we encode the representations of claim image v^c and the evidence image v^e and the corresponding caption t^e . Subsequently, to select the most relevant sample for the claim image, we measure the similarity between the claim image and each evidence image, considering both the claim-evidence image similarity and the claim-evidence text similarity. This selection can be expressed as:

$$\arg \max_{i,j} \left(\frac{e^{\langle v_i, t_j \rangle / \tau}}{\sum_k e^{\langle v_i, t_k \rangle / \tau}} + \frac{e^{\langle t_i, v_j \rangle / \tau}}{\sum_k e^{\langle t_i, v_k \rangle / \tau}} \right) \quad (3)$$

Based on the ranking of the sum of the two similarity scores, we select the top few evidence images that best align with the claim image semantics.

3.3 Knowledge-enhanced Retrieval Module

To enhance textual evidence, FACTCOMPASS introduces a Knowledge-enhanced Retrieval Module, which consists of the SearchAgent for external knowledge retrieval and the RefineAgent for knowledge refinement within documents.

SearchAgent. Claim images often encode rich background knowledge that supports fact-checking by providing context and improving interpretability. We first extract visual entities E_i —such as people and locations—using the Google Entity Detection API and YOLO.

These entities are then enriched through two stages: first, they are linked to the DBpedia knowledge graph to retrieve higher-order semantic information, including one-hop neighboring entities, which are converted into token sequences using a pre-trained model; second, we query Wikidata knowledge base to retrieve concise, entity-specific descriptions, which are contextualized into complete background sentences to support claim understanding in the news domain.

RefineAgent. The RefineAgent processes complex textual evidence $\{T_i^D\}$ retrieved from the internet, minimizing information loss and redundancy. Retrieved documents are divided into fragments, where short ones (1-2 sentences) are treated as a single unit, and longer ones are split into smaller units. Each fragment is converted into a feature vector, and relevance scores are computed to filter out redundant information. The relevant fragments are defined as the set $\hat{T}^D$, and the remaining fragments are merged sequentially to form the final evidence text.

$$\hat{T}^D = \bigcup_{S_i \in \{S_j | \max_{j \neq i} \text{Cos}(\mathbf{v}(S_i), \mathbf{v}(S_j)) \leq \theta\}} S_i \quad (4)$$

where S_i denotes the i -th knowledge fragment, $\mathbf{v}(S_i)$ is its vector representation obtained using BERT [8], and $\text{Cos}(\mathbf{v}(S_i), \mathbf{v}(S_j))$ calculates the cosine similarity between fragments S_i and S_j . The threshold θ filters out redundant information.

3.4 LLM Training

As shown in Figure 3, our pipeline includes three key stages:

Cold Start. Existing MLLMs struggle with misinformation reasoning due to limited exposure to reward modeling during pretraining.

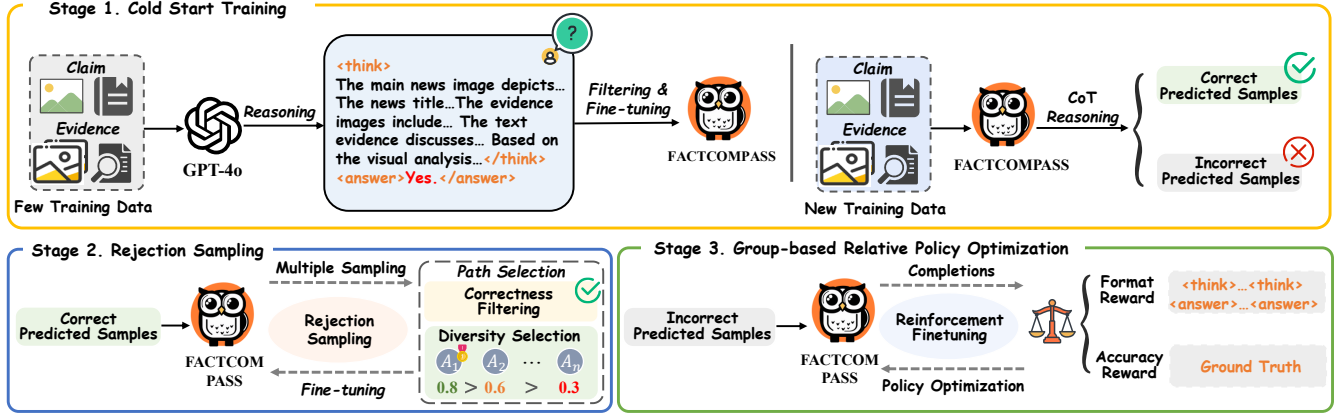


Figure 3: The three-stage self-refining reinforcement fine-tuning strategy.

To address this issue, we introduce a cold-start training phase, where a small amount of high-quality Chain-of-Thought (CoT) reward data distilled from GPT-4o is used to explicitly teach the model the required reasoning format and structure. Specifically, we first prepare a set of training data, each consisting of a multimodal claim, multimodal evidence, and a corresponding question prompt. These samples are then input into GPT-4o to generate detailed long-chain reasoning processes and the final answers. From the generated samples, we retain only those whose reasoning trajectories lead to the correct answers. All reasoning processes are enclosed within `<think>` tags. The training process is driven by minimizing the cross-entropy loss between the model’s predicted output and the correct answers. The cross-entropy loss $\mathcal{L}_{CE}$ is defined as:

$$\mathcal{L}_{CE} = - \sum_{i=1}^N p_{\text{true}}(y_i) \log p_{\text{pred}}(y_i) \quad (5)$$

where $p_{\text{true}}(y_i)$ is the true probability distribution of the correct answer, and $p_{\text{pred}}(y_i)$ is the predicted probability distribution for each sample y_i . After the cold-start training phase, we prompt the model to generalize CoT reasoning across the remaining fact-checking training data, leveraging the prior knowledge acquired during initialization.

Rejection Sampling. After Cold Start training, we further guide and expand the model’s reasoning capabilities in fact-checking. We apply rejection sampling to reinforce the learning of correct fact-enhanced reasoning patterns. Specifically, We retain only those samples whose reasoning aligns with the correct final answer and apply rejection sampling to filter out incorrect or low-quality reasoning. For correct samples, we sample with varying temperatures and select the preferred reasoning via manual diversity evaluation. This process focuses the training distribution on accurate reasoning patterns and enhances the model’s generalization ability across different visual domains.

Group Relative Policy Optimization. We apply GRPO to fine-tune the model’s policy without relying on an explicit value function or a separate critic network. GRPO estimates the relative quality of responses within a group by comparing the outputs generated under the current policy. Given an input, the model samples a

set of candidate outputs $\{y_1, y_2, \dots, y_G\}$, each assigned a reward based on its correctness or alignment with human preferences. The normalized advantage for each response y_i is computed as:

$$\mathcal{L}_{GRPO} = \frac{1}{G} \sum_{i=1}^G \left(\min \left(\frac{\pi_{\theta}(y_i)}{\pi_{\text{old}}(y_i)} A_i, \text{clip} \left(\frac{\pi_{\theta}(y_i)}{\pi_{\text{old}}(y_i)}, 1 - \epsilon, 1 + \epsilon \right) A_i \right) - \beta \mathbb{D}_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \right) \quad (6)$$

Here, π_{θ} represents the current policy, π_{old} is the sampling policy used for generating the responses, and π_{ref} is a fixed reference model. In our setup, each candidate y_i corresponds either to a model-generated output y_{pred} or a human-annotated label y_{label} , allowing us to optimize both relative preferences and ground-truth accuracy. In reinforcement learning, rewards provide the training signals that guide optimization. To train FACTCOMPASS, we design a rule-based reward system with two main components: Accuracy Reward and Format Reward. These rewards are aimed at ensuring both the quality and accuracy of model responses: (1) Accuracy Reward: The accuracy reward r_{acc} is used to evaluate the correctness of the model’s output. It is computed by comparing the model’s predicted class with the true class, encouraging the model to match the true label. This is formalized as:

$$r_{\text{acc}} = \begin{cases} 1, & \text{if model's prediction is correct} \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

(2) Format Reward: In addition to accuracy, we incorporate a format reward r_{format} to ensure that the model encapsulates its reasoning process within the proper output structure, specifically requiring reasoning to be enclosed within the `<think>` and `</think>` tags.

The final reward function $R(q, y)$ combines these two rewards and can be expressed as:

$$R(q, y) = r_{\text{acc}} + r_{\text{format}} \quad (8)$$

Table 1: Accuracy (%) comparison with state-of-the-art methods on three English datasets. News denotes the NewsCLIP-pings dataset. Values marked with * are reported from previous papers.

Methods	News	OOC-MFC	VERITE
BERT	61.1*	68.9	60.3
CLIP	66.0*	72.7	63.6
CCN	84.7*	84.9	79.3
SNIFFER	88.4*	76.3	74.4
RED-DOT	90.3*	85.3	76.9*
AITR	93.3*	88.1	82.7*
DEFAME	86.6	89.0	79.8*
InternVL-2.5	74.7	70.1	66.6
Qwen2.5-VL	79.3	75.5	70.0
Gemini2-thinking	80.8	73.4	70.6
GPT-4o	82.2	84.4*	73.6*
Grok-4	84.5	86.9	72.7
FACTCOMPASS	92.2	90.2	84.1

Table 2: Accuracy (%) comparison with state-of-the-art methods on two Chinese datasets.

Methods	CCMF	SCMF
CCN	79.3*	84.9*
Triple-Check	83.2*	81.3*
ESNet	86.2*	84.9*
InternVL-2.5	63.3	60.6
Qwen2.5-VL	78.8	77.1
Gemini2-thinking	76.4	79.6
GPT-4o	79.9	82.7
Grok-4	82.1	83.2
FACTCOMPASS	88.4	87.7

Table 3: Evaluation of the influence of different components of FACTCOMPASS on the OOC-MFC dataset.

Methods	Acc	Prec	Rec	F1
w/o Evidence Image	85.8	86.3	86.2	86.3
w/o Semantic-enhance	86.5	87.0	86.9	86.9
w/o RefineAgent	84.5	85.7	85.0	85.3
w/o Knowledge Graph	87.4	87.4	87.6	87.5
w/o Knowledge Base	88.4	88.3	88.4	88.4
w/o Cold Start	79.9	82.4	80.8	81.6
w/o Reject	88.7	88.9	88.5	88.7
w/o GRPO	87.2	87.1	87.3	87.2
FACTCOMPASS	90.2	87.5	88.4	88.0

4 Experiments

4.1 Experimental Settings

Dataset. Due to the limitations of available training data, we formulate our task as a binary classification problem. We evaluate the proposed method on three widely used English benchmark datasets—*NewsCLIPpings*[22], *OOC-MFC*[39], and *VERITE*[25]—as

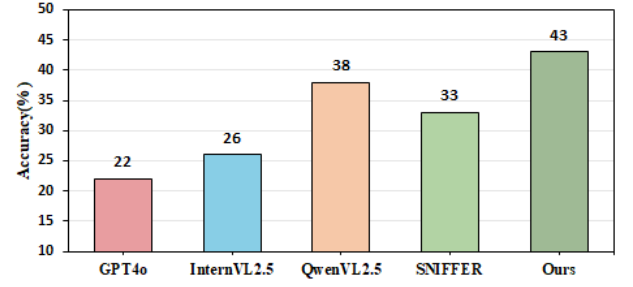


Figure 4: The impact of explainability from five models on GPT-4o review accuracy.

well as the Chinese datasets: *CCMF* and *SCMF*[48]. (1) *NewsCLIP-pings* is designed to detect mismatches between images and captions. Its samples are drawn directly from real news content in the *VisualNews* corpus [20], where both the images and captions remain unaltered, though some are mismatched. We follow the standard protocol by using the provided *balanced* subset with corresponding evidence, treating the task as binary classification and reporting accuracy (Acc) as the evaluation metric. Evidence retrieval and augmentation follow the method described in [1]. (2) *OOC-MFC*, from the *MFC-Bench* benchmark [39], evaluates a model’s ability to detect image-text inconsistencies in context. Although both modalities are individually accurate, their semantic alignment may be incorrect. (3) Furthermore, we evaluate our model on the *VERITE* benchmark [25], which includes 1,000 annotated samples: 338 truthful pairs, 338 miscaptioned images, and 324 out-of-context instances. Following the OOD-CV protocol, we use the external evidence provided by [27], and report the “True vs OOC” accuracy along with standard deviation. (4) Finally, we evaluate our model on the *CMFC* benchmark, a large-scale Chinese multimodal fact-checking dataset comprising 46K claims. It includes two subsets: *CCMF* and *SCMF*. The dataset spans diverse domains such as politics, health, science, and culture. We also review several related benchmarks excluded from our evaluation due to task or format differences. *AVerTeC* [32] contains 500 text-based claims labeled as *Supported*, *Refuted*, *Not Enough Information (NEI)*, and *Conflicting/Cherry-picking (C/CP)*, focusing on claim–evidence reasoning without a training set. *Factify* [23] includes 50K multimodal claims across five fine-grained classes, emphasizing detailed evidence relations beyond binary detection. *MOCHeg* [45] provides 15,601 text-only claims with textual and visual evidence for retrieval, verification, and explanation tasks, thus not directly comparable. *CLAIMREVIEW2024+* [5] introduces 300 recent fact-checking claims (160 unimodal, 140 multimodal) labeled as *Supported*, *Refuted*, *Misleading*, or *NEI*, but its small scale and four-way setup differ from our binary evaluation.

Baselines. We compare our method against a diverse set of multimodal misinformation detection and general-purpose vision-language baselines. BERT [8] serves as a strong text-only baseline using bidirectional transformers for contextual representation learning. CLIP [30] aligns image-text pairs via contrastive learning and is widely adopted as a multimodal backbone. CCN [1] extends CLIP to evaluate intra- and cross-modal consistency between claims

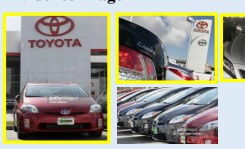
 Text: A Turkish man stands next to a bank window that was damaged during riots after antigovernment demonstrations on June 5 2013 in Istanbul Turkey GT: Fake Evidence image  SearchAgent KG: England, a country located in the United Kingdom, has its capital in London. .. The official language is English, KB: Bank : financial institution that accepts deposits RefineAgent US justice department investigates Iran allegations. Standard Chartered bank accused of scheming with Iran to hide transactions. About 60,000 transactions were involved between 2001 and 2007 and the bank faces losing its ability to trade in the US if the allegations are proven. Discussions continue and the bank said it was ... Prediction <think>The main news image shows an office environment with flags visible through a glass window, which does not align with the claim in the title about a Turkish man standing next to a damaged bank window during riots. The evidence images depict scenes of protests, graffiti, and damage, including a man standing near a damaged window with graffiti, a bus with graffiti, and a bakery scene...</think><answer>Yes</answer>	 Text: A brand new Toyota Camry sits on the sales lot at City Toyota in Daly City Calif in this 2010 file photo GT: Fake Evidence image  SearchAgent KG: Drag racing is a motorsport genre that originated in the United States. It was founded in 1950 and is... KB: Santa Pod Raceway : Europe's first permanent drag racing venue for 1/4 and 1/8 mile racing. RefineAgent The Festival of Power drag racing meet at Santa Pod Raceway Northampton UK Pro Modified Class Elimination round burn out at the start line. Syrians walk past a burning motorcycle at the site of a car bomb explosion in the northern Syrian Kurdish town of Tal Abyad, on the border with... Drag Race Start Stock-Fotos und Bilder - Getty Images. Santa Pod Photos and... Prediction <think>The main news image depicts a drag racing car at Santa Pod Raceway, which is unrelated to the claim in the news title about a brand new Toyota Camry on a sales lot in Daly City, California. However, the main image is clearly manipulated... The text-based evidence does not provide relevant information to support the main image's connection to the claim, further confirming manipulation.</think><answer>Yes</answer>	 Text: Passengers are stranded outside the Guangzhou railway station in China s Guangdong province on Tuesday GT: Fake Evidence image  SearchAgent KG: Hong Kong protests started on March 31, 2019, and continued until December 13, 2019. .. KB: Hong Kong.city and special administrative region of China RefineAgent Under the umbrellas: what do Hong Kong's protesters want from China. High-school demonstrators hold signs during a protest outside the headquarters of Legislative Council in Hong Kong. The protests were predominantly led by young people. The placards read "Hope comes from the people, change begins from struggle" and "peaceful struggle, no compromises"... Prediction <think>The main news image depicts individuals holding protest signs, which is inconsistent with the claim that passengers are stranded outside Guangzhou railway station in Guangdong province. The evidence images... Additionally, the text evidence references Hong Kong's "Umbrella Protests," which are unrelated to the situation described in the news title. This suggests the main new...</think><answer>Yes</answer>	 Text: The memorial is next to the People s Palace in Glasgow GT: Real Evidence image  SearchAgent KG: Glasgow is a city located in Scotland. It has a population of approximately 598,830 people. Glasgow has an area of 175.5 km²... KB: Glasgow .city in Scotland, United Kingdom. RefineAgent People's Palace at Glasgow Green, tells the story of the people of Glasgow. A favourite dog walking spot for myself and Lucy. Glasgow City, Glasgow Green, UK Culture, Places In Scotland, Glasgow Scotland, Highlands Scotland, Countries To Visit, Green Park, Earth... Prediction <think>The main news image depicts a sunset over a body of water, with a distant landmass visible on the horizon. This aligns with the claim that the Turkish mainland is visible from Lesbos, Greece, during sunrise or sunset. The evidence images include scenes of a beach with life... The text evidence discusses sunsets and beaches in general but does not provide specific geographical or scientific...</think><answer>No</answer>
(a)	(b)	(c)	(d)

Figure 5: Four examples of FACTCOMPASS making correct predictions. KG represents the Knowledge Graph, KB represents the Knowledge Base. The selected images for retrieval are highlighted with yellow boxes.

and retrieved evidence. SNIFFER [29], built upon InstructBLIP, performs domain-specific instruction tuning for interpretable out-of-context misinformation detection. InternVL-2.5 [7] integrates a large ViT vision tower and bilingual LLM for bidirectional multimodal fusion. Triple-Check [11] employs a three-stage verification pipeline combining internal consistency analysis, external evidence retrieval, and unified reasoning. Gemini2-thinking [36] and GPT-4o [24] are advanced native multimodal transformers capable of unified text-image reasoning, while Grok-4 [41] offers strong general-purpose reasoning capabilities. DEFAME [5] performs zero-shot six-stage fact-checking via tool invocation and structured justification generation. Qwen2.5-VL [4] enhances vision encoding through ViT-transformer interaction and multimodal pretraining. Finally, AITR [27] and RED-DOT [26] model evidence similarity and relevance-veracity separation via attention-guided fusion and joint optimization. Together, these baselines encompass a broad spectrum of multimodal reasoning and retrieval paradigms for comprehensive comparison.

Implementation Details. All core experiments were conducted three times, and the mean values are reported. We adopt Qwen2.5-VL as the base MLLM, and the training process consists of three stages. DBpedia [2] serves as the background knowledge graph. All experiments are implemented in PyTorch and executed on four NVIDIA Tesla H100 GPUs. In the *Semantic-enhanced Retrieval Module*, the number of evidence images is set to 3; in the *SearchAgent*, the pre-trained model for converting graph triples into token sequences is gpt4o-mini; and in the *RefineAgent*, the similarity threshold is 0.7. In Stage 1 (Cold Start), we initialize training with 20K high-quality

GPT-4o-generated samples from the NewsCLippings dataset, evenly split between *real* and *fake* claims. The visual encoder is frozen, and LoRA is applied for efficient fine-tuning with $\text{lora}_r = 128$, $\text{lora}_\alpha = 256$, and $1r = 2e-5$; the multimodal projector uses a learning rate of $2e-6$. Training runs for 2 epochs with a batch size of 2 using AdamW and cosine annealing scheduling. In Stage 2 (Rejection Sampling), we retain model-generated responses with accurate reasoning and manually select the best explanations from three temperature-varied outputs (0–1), yielding 20K refined samples. LoRA is reused with $\text{lora}_r = 64$, $\text{lora}_\alpha = 128$, and $1r = 2e-5$, while keeping the projector learning rate at $2e-6$ and the visual encoder frozen. In Stage 3 (GRPO), Group-based Relative Policy Optimization is applied with a verifiability-based reward. The model samples eight candidate outputs per input and is trained for 2 epochs on 20K news samples with $1r = 2e-5$. Final results are reported as the average over three independent runs to ensure stability and reliability.

4.2 Comparison to SOTA

(1) As shown in Table 1 on the three English datasets, early models such as CLIP [30], and BERT [9] demonstrate limited robustness and shallow semantic alignment. Recent approaches like CCN [1] improve performance by integrating external knowledge and symbolic reasoning, while SNIFFER [29] provides explainable outputs but lacks fine-grained evidence-level modeling. RED-DOT [26] separates evidence relevance from veracity prediction in a unified transformer. AITR [27] introduces a simple yet effective baseline

for multimodal similarity evaluation. Closed-source models including GPT-4o [24], Gemini2-thinking [36] and Grok-4 [41] achieve the strongest overall results, highlighting their powerful reasoning capabilities. Meanwhile, open-source models such as Qwen2.5-VL also perform competitively, underscoring the growing potential of open MLLMs. (2) As shown in Table 2 on the two Chinese datasets, our method achieves the best overall performance on both benchmarks. Qwen2.5-VL exhibits strong zero-shot results due to its superior Chinese understanding ability. Triple-Check [11] follows a three-stage pipeline—internal consistency checking with lightweight attention. ESCNet combines entity-aware modeling with cross-modal stance verification and obtains competitive results with a lightweight attention mechanism. Compared to these, FACTCOMPASS integrates semantic and knowledge-enhanced retrieval with a self-refining reinforcement strategy to improve evidence alignment, knowledge-level verification, and robust reasoning against misinformation.

4.3 Explainability Analysis

To further assess effectiveness in revealing misinformation, we selected a subset of 100 challenging multimodal claims that all five models in Figure 4 failed to classify correctly—comprising 50 real claims incorrectly predicted as fake, and 50 fake claims incorrectly predicted as real, resulting in 0% accuracy across the board. For each instance, we retained the model-generated explanation. To simulate human fact-checking, we designed a prompt that inputs each claim and its explanation into GPT-4o, asking it to make a secondary judgment (real or fake). As shown in Figure 4, explanations generated by FACTCOMPASS significantly improved the secondary review accuracy, while GPT-4o’s own explanations often amplified hallucinations and biases. Additional analysis can be found in *Supplementary Materials*.

4.4 Ablation Studies

We conduct ablation studies to evaluate the impact of each module in FACTCOMPASS, as shown in Table 3. (1) For the Semantic-enhanced Retrieval Module, removing either the evidence image token (*w/o Evidence Image*) or the retrieval module (*w/o Retrieval*, replaced with CLIP similarity) leads to notable performance drops, highlighting the module’s importance for aligning visual evidence. (2) For the Knowledge-enhanced Retrieval Module, omitting the RefineAgent degrades performance, confirming its role in extracting concise, relevant information from textual evidence. Removing either the knowledge graph or knowledge base from the SearchAgent also reduces accuracy, demonstrating the benefit of multi-level external knowledge. (3) Finally, ablating components of the self-refining reinforcement fine-tuning strategy—*w/o Cold Start*, *w/o Reject*, and *w/o GRPO*—shows that each stage plays a critical role in gradually enhancing the model’s reasoning ability.

4.5 Qualitative Evaluation

Figure 5 presents four examples that were correctly identified, with enhanced evidence text and evidence image provided by the two retrieval modules. By examining these examples, we observe that our semantic-enhanced retrieval module can effectively uncover

evidence images semantically similar to the claim image and refine the associated evidence text. Furthermore, the SearchAgent effectively captures key knowledge embedded in the claim image, providing valuable background information for the news content. For instance, in example (c), the news content is relatively obscure, but by retrieving the entity ‘Hong Kong’ and its contextual description, the model fully comprehends the news, aiding in fake claim detection. In example (d), By converting news background knowledge into textual modality, the model improves its understanding of world locations like ‘Glasgow’ in relation to the claim image, thereby enhancing the multimodal fact-checking process. Furthermore, our model not only produces accurate predictions but also generates coherent and interpretable reasoning chains.

5 Conclusion

We propose FACTCOMPASS, a comprehensive and unified multimodal fact-checking framework designed to enhance both reasoning depth and retrieval precision through three synergistic components: semantic-enhanced evidence image retrieval, knowledge-enriched textual refinement, and a self-refining reinforcement fine-tuning strategy. By jointly optimizing semantic understanding, knowledge grounding, and reasoning adaptability, FACTCOMPASS achieves more accurate and interpretable multimodal fact verification. Extensive experiments further validate its superior performance, robust reasoning ability, and high interpretability in misinformation detection and related applications.

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Claim Text	Claim Image	SNIFFER	GPT4o	FACTCOMPASS
#1: Oil tanker being moved into position at Fawley Refinery in Southampton		On the one hand, the caption refers to an oil tanker being moved into position, while the image shows a tugboat assisting with the maneuvering. On the other hand, the image lacks clear indicators of the Fawley Refinery, making the location reference unclear. Therefore, the image seems to be mismatched with the caption. ✗	The image depicts a busy industrial scene with large ships and towering refinery infrastructure in the background, resembling an oil refinery or port area, which might be similar to the process of moving an oil tanker. However, the image doesn't explicitly show the Fawley Refinery in Southampton, nor does it confirm the presence of an oil tanker. Instead, it focuses on a tugboat maneuvering a large vessel near an industrial complex. Therefore, while the general activity and setting suggest a port or refinery, the image does not perfectly align with the specific location or description provided in the title. ✗	<think>The main news image shows a tugboat labeled "THRAX" assisting an oil tanker near a refinery, which aligns with the claim in the title. The evidence images are consistent with the scene depicted in the main image, showing similar angles and elements such as the tugboat, oil tanker, and refinery infrastructure. There are no indications of manipulation or inconsistency between the images and the claim. Despite the failure to fetch text-based evidence from the knowledge base and graph, the visual evidence supports the claim.</think> ✓
#2: A SWAT team responded to a report of multiple people shot in Belfair Wash Feb 26 2016		On the one hand, the caption refers to a SWAT team responding to a report of multiple people shot, indicating a serious crime scene and potentially a life-threatening situation. On the other hand, the image shows law enforcement vehicles parked on a forested road, but it doesn't clearly depict a SWAT team in action or any immediate signs of a shooting incident. ✗	The image depicts law enforcement vehicles lined up on a forested road, suggesting a tactical or investigative operation. However, the scene does not explicitly show a SWAT team actively responding to a reported shooting in Belfair, Washington, on February 26, 2016. While the image suggests a law enforcement presence, it lacks the specific context of a shooting or the direct involvement of SWAT officers, making it inconsistent with the description provided. ✗	<think>The news title claims that a SWAT team responded to a report of multiple people shot in Belfair, Washington, on February 26, 2016. The main news image shows a rural area with police vehicles and personnel, consistent with a law enforcement response. The evidence provided includes multiple credible sources (e.g., Reuters, NDTV, The Himalayan Times) reporting on a murder-suicide incident in Belfair, Washington, where five people, including the suspect, were found dead. There is no indication that the claim has been manipulated, as the visual and textual evidence corroborates the event described in the title.</think> ✓
#3: Rittenhouse Square in Philadelphia a city known for its history and culture		On the one hand, the caption refers to "Rittenhouse Square in Philadelphia," a city known for its history and culture. On the other hand, the image, while showing a street scene with American flags and tall buildings, reflects the urban vibe and cultural ambiance often associated with major cities like Philadelphia. ✗	The image shows a cityscape with American flags hanging from buildings, likely in an urban setting. While the title mentions Rittenhouse Square in Philadelphia, the scene depicted in the image seems to reflect a more general urban atmosphere, possibly a street in a city with tall buildings and flags displayed. This could be interpreted as a representation of a proud, vibrant urban area, likely in a city known for its history and culture, as the title suggests. ✗	<think>The main news image depicts a street scene with American flags and skyscrapers, which is characteristic of New York City, specifically near the Chrysler Building. This does not align with the claim that the image represents Rittenhouse Square in Philadelphia, which is known for its park-like setting and historical architecture. The evidence images clearly show Rittenhouse Square as a park with trees, benches, and open spaces, consistent with its reputation as a cultural and historical landmark in Philadelphia. The mismatch between the main image and the claim indicates manipulation.</think> ✓
#4: Ted Cruz greets supporters on April 5 2016 in Milwaukee		On the one hand, the caption refers to Ted Cruz greeting supporters on April 5, 2016, in Milwaukee, which aligns with the image showing Ted Cruz speaking to an audience, likely at a campaign event. On the other hand, the image features Cruz in front of a backdrop with his name and the word "TED," which is consistent with campaign visuals. The setting appears to be a rally, and Cruz is actively engaging with the crowd, matching the context of him greeting supporters. ✗	The image features Ted Cruz, a U.S. politician and senator, speaking to a group of supporters. The background displays a banner with the word "TED" in large letters, accompanied by the words "COURAGEOUS," "CONSERVATIVE," and "CONSISTENT," which aligns with his campaign slogan. Cruz is seen engaging with the crowd, holding a microphone in one hand while raising his other hand in a gesture, likely to emphasize his speech. ✗	<think>The main news image shows Senator Ted Cruz walking in a hallway, which does not align with the claim that he is speaking during the 2016 Republican National Convention. The evidence images depict him in settings that match the convention, such as speaking at a podium with a patriotic backdrop. This discrepancy suggests the claim is manipulated, as the main image does not correspond to the event described in the title.</think> ✓

Figure 6: The explainable outputs of three models are as follows: Case 1 and Case 2 are real claims mistakenly classified as fake by GPT4o and SNIFFER, while Case 3 and Case 4 are fake claims mistakenly classified as real by GPT4o and SNIFFER.

Query Claim Image	Evidence Image with the caption	Inference Score
 <i>Real</i>	 'Doulton Fountain, Glasgow'  The memorial is next to the People's Palace in Glasgow  Nelson Monument, Glasgow	Image-Text Inference Score 0.945 0.998 0.881 Evidence1 Evidence2 Evidence3 Image-Image Inference Score 0.991 0.982 0.995 Evidence1 Evidence2 Evidence3
 <i>Fake</i>	 China Guangdong Guangzhou Province City Guangzhou Railway Station High Resolution Stock Photography and Images  Massive queues outside Guangzhou railway station in Guangzhou, southern China's Guangdong province, on February 2nd, 2016.  Tens of thousands of passengers were stranded outside Guangzhou railway station in Guangzhou.	Image-Text Inference Score 0.721 0.697 0.655 Evidence1 Evidence2 Evidence3 Image-Image Inference Score 0.799 0.608 0.659 Evidence1 Evidence2 Evidence3

Figure 7: Visualization of the inference process of the Semantic-enhanced Retrieval Module. The evidence images, along with their captions, are arranged from top to bottom as Evidence 1, Evidence 2, and Evidence 3.

A Pseudo Code of FACTCOMPASS

In Algorithm 1, we summarize the training and inference processes of the Semantic-enhanced Retrieval Module. The training phase involves optimizing a Fact NCE contrastive loss using claim-image, evidence-image, and cross-image-text pairs. During inference, the model evaluates the similarity between claim images and candidate evidence images by considering both image and caption alignment, selecting the most relevant evidence based on semantic consistency.

B More Experiments

B.1 Explainability Analysis

We present four examples from the datasets in Figure 6 that were mistakenly classified by GPT4o and SNIFFER. Case 1 and Case 2 are real claims mistakenly classified as fake, while Case 3 and Case 4 are fake claims mistakenly classified as real. We observe that both SNIFFER and GPT-4o tend to forcefully output some entity inconsistencies. For instance, in Case 1, SNIFFER generates the hallucination,

You are a **professional multimodal fact-checker**. Your task is to assess the accuracy of a given claim by analyzing both textual and visual evidence. You will be provided with the following three inputs:

1. **Claim Text**: A textual statement or description.
2. **Claim Image**: An image associated with the claim text.
3. **Evidence**: The description of the visual elements in the image, such as objects, people, or landmarks, may contain errors.
4. **Explainability**: A detailed explanation or interpretation that may provide context or clarification regarding the claim text or image. **The content is provided for reference only and may contain inaccuracies.**

Your objective is to determine whether the claim is "real" or "fake" based on these inputs. Consider the following when making your assessment:

- **Consistency**: Does the claim text align with the content of the image? Are there any contradictions between the two?
- **Relevance of Explainability**: Does the explainability provide additional context or clarification that helps confirm or disprove the claim?
- **Contextual Accuracy**: Are there any factual inaccuracies in the claim text, image, or explanation? Use these guidelines to classify the claim as either "real" or "fake":
 - If the claim text and image show significant inconsistencies, and the explainability supports or highlights this, the claim is "fake".
 - If the claim text and image are aligned and supported by the explainability, the claim is "real".

Example:

Claim Text: "Rittenhouse Square in Philadelphia, a city known for its history and culture."

Claim Image: [image path]

Evidence: "American flag, skyscrapers, streets"

Explainability: "American flags hang between towering skyscrapers, framing the bustling cityscape."

Output: Fake

Reasoning: The claim text suggests a historical and cultural location, but the image shows a modern, skyscraper-filled cityscape, which contradicts the cultural and historical character of Rittenhouse Square. This inconsistency, supported by the explanation, indicates that the claim is likely fake.

Input Format:

- **Claim Text**: [insert claim text here]
 - **Claim Image**: [insert image path here]
 - **Evidence**: [insert evidence image and evidence text here]
 - **Explainability**: [insert explainability here]
- Output Format:**
- "real" or "fake" (based on your judgment)

Figure 8: The prompt we use to instruct GPT-4o to mimic human reviewers for a secondary verification process.

You are an expert in multimodal fact-checking. Given the news title, news image, and optional evidence, determine if the claim is manipulated.

News Title: {caption}

Instructions:

- Analyze the first news image <image>.
- The next images are evidence <image><image><image>.
- Use the text-based evidence below to reason (they may contain errors, use judgment).

Evidence Text:

{evidence_block}

Respond in the following format:

<think>Reasoning goes here</think><answer>Yes. or No.</answer>

Example:

Claim Text: "Saudi troops cheer as they ride at the back of an army truck in the southwestern province of Jizan near the border with Yemen."

Claim Image: [image path]

Evidence Image: [image path1], [image path2], [image path3]

Evidence Text: "Israel : country in Western Asia | Prime minister : most senior minister of cabinet in the executive branch of government in a parliamentary system and ..."

Output: <think>The main news image shows soldiers wearing helmets and military gear, which is inconsistent with the description of Saudi troops in Jizan. The helmets and gear resemble those used by Israeli soldiers, as seen in the context of the Israeli-Palestinian conflict. The evidence images depict soldiers in different uniforms and vehicles, more consistent with Saudi military. The text evidence discusses unrelated geopolitical issues, primarily focusing on Israel and Palestine, which further suggests the main image is not related to Saudi troops. The ground truth label "Yes" indicates the claim is manipulated, supporting the conclusion that the image does not depict Saudi troops in Jizan.</think><answer>Yes.</answer>

Figure 9: The prompt we use to instruct GPT-4o to generate reasoning data during Cold Start Training.

Algorithm 1: Semantic-enhanced Retrieval Module: Training and Inference

Input: Mini-batch of N samples. Each sample includes claim image-text pairs (v_i^c, t_i^c) , evidence image-text pairs (v_i^e, t_i^e) , and cross-image-text pairs (v_i^c, t_i^e) , (v_i^e, t_i^c) .

Output: Optimized retrieval model.

// Training phase

for each sample in the mini-batch do

 Compute image-to-text NCE loss $\mathcal{L}_{i2t}$ // as defined in Eq. (1);

 Define positive pair set:

$\mathcal{M}_i = \{(v_i^c, t_i^c), (v_i^e, t_i^e), (v_i^c, t_i^e), (v_i^e, t_i^c)\}$;

 Compute text-to-image NCE loss $\mathcal{L}_{t2i}$ // as defined in Eq. (2);

 Compute total loss $\mathcal{L}_{\text{total}} = \mathcal{L}_{i2t} + \mathcal{L}_{t2i}$;

 Update model parameters to minimize $\mathcal{L}_{\text{total}}$;

end

// Inference phase

for each claim image v^c and candidate evidence image v^e do

 Encode v^c , v^e , and evidence caption t^e ;

 Compute image-caption similarity scores // as defined in Eq. (3);

 Select top-ranked evidence images based on similarity;

end

return Optimized model for evidence image matching in the news domain.;

'oil tanker being moved into position, while the image shows a tug-boat assisting with the maneuvering.' Such hallucinations in the explainable outputs can mislead human fact-checkers. In contrast, FACTCOMPASS delivers accurate and logically structured reasoning chains, empowering the model to generate interpretable and domain-aware fact-checking results beyond surface-level matching.

We illustrate two tailored prompts used to instruct GPT-4o in Figure 8 and Figure 9. Specifically, the prompt in Figure 8 is designed to simulate human reviewer behavior for secondary verification, enabling GPT-4o to assess claim-image pairs from a fact-checking perspective. Meanwhile, the prompt in Figure 9 is crafted to elicit high-quality reasoning traces during the Cold Start Training Stage.

B.2 Qualitative Evaluation

Figure 7 illustrates the inference matching process of the Semantic-enhanced Retrieval Module, presenting the top three evidence image-caption pairs with the highest composite scores. For fake claims, both the Image-Text and Image-Image Inference Scores are relatively low and uniformly distributed, leading the retrieval process to select semantically similar but irrelevant pairs (e.g., "China Guangdong Guangzhou Province City"). In contrast, for pristine claims, the module successfully identifies evidence images that are semantically consistent with both the claim image and text (e.g., "Nelson Monument, Glasgow"). These results demonstrate that, through contrastive learning and multimodal training, the Semantic-enhanced Retrieval Module effectively captures semantic coherence across modalities, enabling accurate and meaningful evidence matching.